

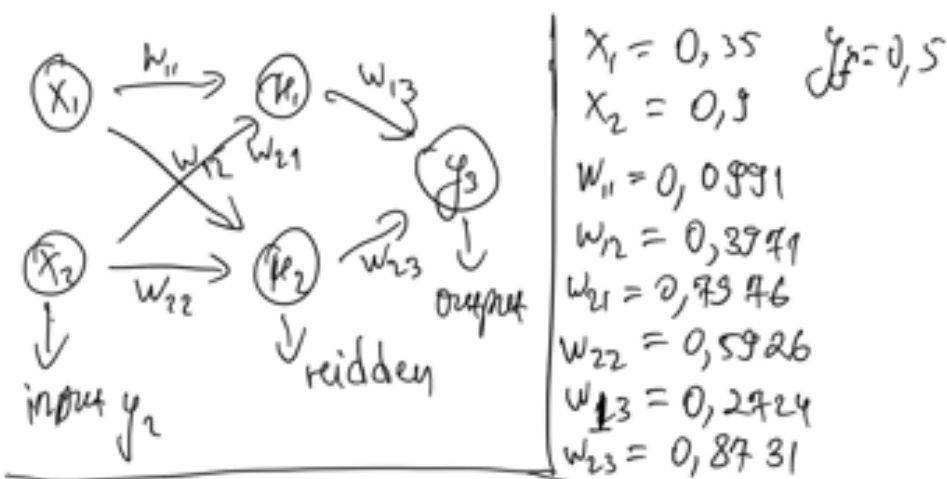


NLP Week 11 Homework: Neural Networks

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NLP homework / Week-11 (Neural Networks)
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$$[h_i = \sum v_{ij} x_j] [f(h_i) = y_i] [f(x) = \frac{1}{1+e^{-x}}]$$

Sigmoid Acene

1st Iteration:

$$\begin{aligned}
 y_1 &= \sum w_{ij} x_i \Rightarrow (x_1 \cdot w_{11}) + (x_2 \cdot w_{21}) = \\
 &= (0,35 \cdot 0,0991) + (0,9 \cdot 0,7946) = \\
 &= 0,034685 + 0,71784 = 0,752525 \approx 0,7525 \\
 f(y_1) &= \frac{1}{1+e^{-0,7525}} \approx \underline{0,6797}
 \end{aligned}$$

$$\begin{aligned}
 y_2 &= (x_1 \cdot w_{12}) + (x_2 \cdot w_{22}) = (0,35 \cdot 0,3971) + \\
 &+ (0,9 \cdot 0,5926) = 0,138985 + 0,53334 = \\
 &= 0,672325 \approx 0,6723
 \end{aligned}$$

$$f(y_2) = \frac{1}{1+e^{-0,672325}} \approx 0,662018$$

$$\begin{aligned}
 y_3 &= (h_1 \cdot w_{13}) + (h_2 \cdot w_{23}) = (0,6797 \cdot 0,2924) + \\
 &+ (0,662018 \cdot 0,8731) \approx \underline{0,7920}
 \end{aligned}$$

$$f(y_3) = \frac{1}{1+e^{-0,7920}} \approx 0,688261$$

$$Error_j = y_1 - y_3 = 0,5 - 0,688261 = -0,188261$$

Backpropagation(1)

$$[\eta = 0,1] \quad [\Delta w_{new} = \eta \cdot \delta_j \cdot y_i + w_{old}]$$

$$\delta_i = \begin{cases} \text{output error} & \text{if } i \text{ is output} \\ \text{propagated error} & \text{if } i \text{ is hidden} \end{cases}$$
$$\delta_i = \begin{cases} y_i(1-y_i)(y_t - y_i) & \text{if } i \text{ is output} \\ y_p(1-y_p)(w_{ij} \cdot \delta_{pj}) & \text{if } i \text{ is hidden} \end{cases}$$

$$\delta_3 = y_3(1-y_3)(y_t - y_3) = 0,6883(1-0,6883) \cdot (0,5 - 0,6883) = 0,6883 \cdot 0,3117 \cdot (-0,1883) = -0,0404$$

$$\delta_2 = y_2(1-y_2)(w_{23} \cdot \delta_3) = 0,6620(1-0,6620) \cdot 0,8731 \cdot (-0,0404) = -0,0075$$

$$\delta_1 = y_1(1-y_1)(w_{13} \cdot \delta_3) = 0,6797(1-0,6797) \cdot (0,2727) \cdot (-0,0404) = -0,0024$$

$$\Delta w_{23} = 0,1 \cdot \delta_3 \cdot y_2 = 0,1 \cdot (-0,0404) \cdot 0,6620 = -0,0027 + w_{old} = (-0,0027) + 0,8751 = 0,8704$$

$$\Delta w_{13} = 0,1 \cdot \delta_3 \cdot y_1 + w_{old} = (0,1 \cdot (-0,0404) \cdot 0,7525) + 0,2727 = 0,2694$$

$$\Delta w_{11} = 0,1 \cdot \delta_1 \cdot x_2 + w_{old} = (0,1 \cdot (-0,0024) \cdot 0,357) + 0,0991 = (-0,000085) + 0,0991 = 0,099015$$

$$\Delta w_{21} = (0,1 \cdot \delta_1 \cdot x_2) + w_{old} = (0,1 \cdot (-0,0024) \cdot 0,797) + 0,7976 = 0,797384$$

$$\Delta w_{22} = (0,1 \cdot \delta_2 \cdot x_2) + w_{old} = (0,1 \cdot (-0,0075) \cdot 0,5) + 0,5920 = 0,591825$$

$$\Delta w_{12} = (0,1 \cdot \delta_2 \cdot x_1) + w_{old} = (0,1 \cdot (-0,0075) \cdot 0,35) + 0,3971 = 0,396225$$

2nd Iteration:

$$y_1 = (x_1 \cdot \Delta w_{11}) + (x_2 \cdot \Delta w_{21}) = (0,35 \cdot 0,099) + (0,9 \cdot 0,7975) = 0,75231$$

$$f(y_1) = \frac{1}{1 + e^{-0,75231}} \approx 0,679682$$

$$y_2 = (x_1 \cdot \Delta w_{12}) + (x_2 \cdot \Delta w_{22}) = (0,35 \cdot 0,396824) + (0,9 \cdot 0,591889) = 0,138887 + 0,5327001 = 0,6716$$

$$f(y_2) \approx 0,661861$$

$$y_3 = (x_1 \cdot \Delta w_{13}) + (x_2 \cdot \Delta w_{23}) = (0,75231 \cdot 0,2694) + (0,6716 \cdot 0,8704) = 0,20267 + 0,58456 = 0,78723 \Rightarrow f(y_3) \approx 0,68723$$

$$\text{Error}_2 = 0,5 - 0,68723 = -0,18723$$

Backpropagation (2):

$$\delta_3 = y_3(1 - y_3)(y_3 - 0,5) = 0,6872(1 - 0,6872) \cdot (0,5 - 0,6872) = 0,6872 \cdot 0,3128 \cdot (-0,1872) = -0,04024$$

$$\delta_2 = y_2(1 - y_2)(\Delta w_{23} \cdot \delta_3) = 0,6619(1 - 0,6619) \cdot (0,8704 \cdot (-0,04024)) = -0,00784$$

$$\delta_1 = y_1(1 - y_1)(\Delta w_{13} \cdot \delta_3) = 0,679682 \cdot (1 - 0,679682) \cdot (0,2694 \cdot (-0,04024)) = -0,00236$$

$$\Delta w_{23}^1 = (0,1 \cdot \delta_3 \cdot y_2) + \Delta w_{23} = (0,1 \cdot (-0,04024) \cdot 0,6619) + 0,8704 = 0,86773$$

$$\Delta w_{13}^1 = (0,1 \cdot \delta_3 \cdot y_1) + \Delta w_{13} = (0,1 \cdot (-0,04024) \cdot 0,679682) + 0,2694 = 0,2667$$

$$\Delta w_{11}^1 = (0,1 \cdot \delta_1 \cdot x_1) + \Delta w_{11} = (0,1 \cdot (-0,00236) \cdot 0,35) + 0,099016 = 0,09853$$

$$\Delta w_{21}^1 = (0,1 \cdot \delta_1 \cdot x_2) + \Delta w_{21} = (0,1 \cdot (-0,00236) \cdot 0,9) + 0,797374 = 0,7972$$

$$\Delta w_{22}^1 = (0,1 \cdot \delta_2 \cdot x_2) + \Delta w_{22} = (0,1 \cdot (-0,00784) \cdot 0,9) + 0,591889 = 0,5898$$

$$\Delta w_{12}^1 = (0,1 \cdot \delta_2 \cdot x_1) + \Delta w_{12} = (0,1 \cdot (-0,00784) \cdot 0,35) + 0,39682 = 0,3965$$

3rd Iteration

$$\begin{aligned} y_1 &= (x_1 \cdot \Delta w_{11}') + (x_2 \cdot \Delta w_{21}') = (0,35 \cdot 0,09893) + \\ &+ (0,9 \cdot 0,1972) = 0,0346255 + 0,17748 = \\ &= 0,2121 \end{aligned}$$

$$f(y_1) = \frac{1}{1 + e^{-0,2121}} \approx \underline{0,6796}$$

$$\begin{aligned} y_2 &= (x_1 \cdot \Delta w_{12}') + (x_2 \cdot \Delta w_{22}') = \\ &= (0,35 \cdot 0,3965) + (0,9 \cdot 0,5848) = \\ &= 0,138775 + 0,52632 = 0,6651 \end{aligned}$$

$$f(y_2) = \frac{1}{1 + e^{-0,6651}} \approx \underline{0,6609}$$

$$\begin{aligned} y_3 &= (h_1 \cdot \Delta w_{13}') + (h_2 \cdot \Delta w_{23}') = (0,4521 \cdot 0,2661) + \\ &+ (0,6651 \cdot 0,86473) = 0,1202338 + 0,57522723 = \\ &= 0,695461 \end{aligned}$$

$$f(y_3) = \frac{1}{1 + e^{-0,695461}} \approx \underline{0,6851}$$

$$\text{Error} = f_2 - y_3 = 0,5 - 0,6851 = \underline{-0,1851}$$

Results:

	1st Iteration	2nd Iteration	3rd Iteration
y_1	0.7525	0.7523	0.7521
$f(y_1)$	0.6797	0.6768	0.6796
y_2	0.6723	0.6716	0.6651
$f(y_2)$	0.6620	0.6619	0.6604
y_3	0.7920	0.7872	0.7773
$f(y_3)$	0.6883	0.6872	0.6851
w_{11}	0.0991	0.0990	0.0989
w_{12}	0.3971	0.3968	0.3965
w_{21}	0.7976	0.7974	0.7972
w_{22}	0.5926	0.5919	0.5848
w_{13}	0.2724	0.2694	0.2667
w_{23}	0.8731	0.8704	0.8677
<i>Error</i>	-0.1883	-0.18723	-0.1851

It can be seen that with the use of Back propagation we were able to adjust the weights in such a way that the value of error decreased with each iteration.